MAHENDRA GURJAR

AI/ML Engineer | LLM Inference, CUDA & Quantization

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SUMMARY

I'm a final-year AI & ML student who learns by building things from scratch. I wrote my own LLM inference engine in C++ and CUDA, re-implemented llama.cpp's quantization in C, and published quantized models on Hugging Face. I like to measure what I build rather than assume it works. I'm looking for an entry-level AI/ML engineering role on a team that trains or serves its own models.

EDUCATION

Bachelor of Engineering, Artificial Intelligence & Machine Learning Expected 2027

L.D. College of Engineering, Ahmedabad

TECHNICAL SKILLS

Languages	Python, C++, C, CUDA, JavaScript
ML / Deep Learning	PyTorch, Hugging Face Transformers, LLMs, Mixture-of-Experts, Fine-tuning (DPO, PPO, GRPO), RAG, LLM Agents
LLM Systems	Inference Optimization, KV Cache, Quantization (GGUF, GPTQ, AWQ), CUDA Kernels, cuBLAS
Tools	Linux, Git, Docker, CMake, FastAPI, Google Colab

PROJECTS

INFERNT – LLM Inference Engine from Scratch (C++, CUDA) GitHub

- Wrote a complete inference engine for Llama-style models in C++ and CUDA, with no PyTorch, ONNX Runtime or TensorRT. It has its own tokenizer and loads Hugging Face checkpoints directly.
- Wrote my own CUDA kernels for attention, RMSNorm, RoPE, softmax and SwiGLU, and used cuBLAS for the matrix multiplications. All GPU memory is set up once, so text generation never stops to allocate.
- Runs TinyLlama-1.1B at about 53 tokens/sec on an NVIDIA T4 GPU. I checked every part against a CPU version to make sure the outputs match.

LLM Quantization – GGML/GGUF Implementation (C, Hugging Face) GitHub

- Studied llama.cpp's source and rebuilt its Q4_0 and Q4_K 4-bit formats in C, then wrote a report on how much space each one really uses (about 4.5 bits per weight).
- Shrunk TinyLlama-1.1B-Chat into GGUF and AWQ versions and published both on Hugging Face.

Mixture-of-Experts Model from Scratch (PyTorch) GitHub

- Built a Mixture-of-Experts layer where each token picks its 2 best experts, and trained an 8-expert model that reached a validation loss of 2.50 (random guessing gives 4.17).
- Tracked how work was shared across experts and found every expert stayed in use, with no dead experts.

Mechanistic Interpretability – Induction Heads in GPT-2 (PyTorch) GitHub

- Found the attention heads that let GPT-2 copy patterns it has already seen in a prompt. Switching off the main one made the model about 45% worse at the task.

OPEN SOURCE & ADDITIONAL WORK

- **MemTrace** (GitHub): An open-source Python library I released that shows why an AI agent forgot or misremembered something.
- **RAG & Fine-tuning:** Built several retrieval pipelines (Corrective RAG, Self-RAG, GraphRAG on Neo4j) and notebooks for fine-tuning models with DPO, PPO and GRPO.